

Data Card

Agricultural Extension RAG Retrieval Dataset

1. Dataset Summary

This dataset is a document-retrieval benchmark for a Retrieval-Augmented Generation (RAG) system that answers smallholder farmers' agronomic questions. The model's input is a single natural-language farmer question (e.g. "How do I manage bacterial leaf blight in rice?"), and its output is a ranked list of the 5 most relevant documents drawn from a fixed knowledge base of 695 short agricultural-extension factsheets. The knowledge base blends two sources: a synthetic backbone of 637 originally authored factsheets (CC0) covering climate risks, pests, diseases, and crop management, and 58 LLM-grounded rewrites of real, open-access CGIAR (CGSpace) advisory documents (CC-BY-4.0). Documents are attributed to realistic extension publishers (FAO, CGIAR, Plantwise, ICRISAT, IITA, AGRA, National Extension Service) and tagged by crop (13 crops) and country context (21 African countries). Relevance is graded on a 0-3 scale across 4,194 query-document judgments covering 308 training queries and 200 held-out test queries, with an average of 4.8 relevant documents per query and deliberate hard negatives (topically similar but incorrect crop/disease combinations) to stress-test retrieval precision. The dataset directly supports the project's goal: helping a RAG pipeline surface trustworthy, regionally relevant extension advice for the kinds of plain-language questions a smallholder farmer would actually ask, evaluated with nDCG@5 against a TF-IDF baseline of 0.551.

Corpus at a glance

Metric	Value
Knowledge base (documents.csv)	695 documents
Training queries (train_queries.csv)	308 queries
Held-out test queries (test_queries.csv)	200 queries
Graded relevance judgments (qrels_train.csv)	4,194 query-document pairs
Avg. relevant docs per training query	4.8
Relevance scale	0 = hard negative, 1 = marginal, 2 = relevant, 3 = perfect match
Crops covered	13 (+ general soil/climate topics)
Countries represented	21 (African smallholder contexts)
Evaluation metric / baseline	nDCG@5; TF-IDF baseline = 0.551

2. Input vs. Output

Model input

A single free-text farmer question in English, e.g. "What should I do about an outbreak of bacterial leaf blight in rice?" or "My common bean shows spots and damage — could it be angular leaf spot?". Questions are phrased the way a farmer would actually ask them, not as keyword search terms.

Model output

A ranked list of the top-5 document IDs from the 695-document knowledge base, best match first. Each retrieved document is a short extension factsheet (title + body text) addressing a specific crop, risk, pest, or disease, e.g. an entry on drought adaptation or bean anthracnose management. The system is a retriever/ranker, not a generator: it does not produce free-text answers, only an ordered shortlist of source documents that a downstream RAG-based answer generator (or extension worker) would draw on.

3. Task Definition (Relevance Grading in lieu of Classification)

This is a ranking/retrieval task rather than a classification task, so there are no output class labels. Instead, each (query, document) pair used for training and evaluation carries a graded relevance label:

- **3 — Perfect match:** the document directly and completely answers the farmer's question.
- **2 — Relevant:** the document is on-topic and useful but not the single best answer.
- **1 — Marginal:** the document is loosely related (e.g. same crop, different issue).
- **0 — Hard negative:** the document is topically adjacent (similar crop/disease/region) but incorrect, included specifically to penalize shallow keyword matching.

These grades were chosen because they mirror how a real extension officer would triage source material — not every partially-related fact sheet is equally useful — and because nDCG@5, the competition's leaderboard metric, requires graded (not binary) relevance to reward ranking the single best document first.

4. Labelling Strategy

Relevance labels were assigned programmatically at dataset construction time rather than by independent human or crowd annotation. Each query was authored as one of several natural-language paraphrase templates tied to a specific crop/issue combination already present in the knowledge base (e.g. "How do I manage X in Y?", "What should I do about an outbreak of X in Y?", "What are the symptoms of X in Y?"), so the correct positive document(s) were known by design rather than inferred after the fact. Hard negatives were then deliberately added by selecting documents that share surface features (same crop, similar disease family, or same region) with the true positives but are not the correct answer, forcing a retriever to reason beyond keyword overlap.

Implication for our project: because labels are template-generated rather than independently verified by domain experts or farmers, our team is treating them as a strong training/evaluation signal for retrieval quality, but not as a substitute for expert review of the underlying agronomic content itself before any advice reaches a real farmer.

5. Data Creation

The 695-document knowledge base was assembled from two clearly distinguished origins (tracked in the *origin* field):

- **Synthetic backbone — 637 documents (91.7%), CC0:** originally authored short factsheets covering climate risks (drought, flooding, heat stress, shifting seasons) and crop-specific pest/disease management across 13 crops, written to reflect realistic regional variation (e.g. the same drought-adaptation topic rewritten for the Sahel, the Guinea savanna, and the humid forest zone) and attributed to representative extension-publisher names for realism.
- **LLM-grounded rewrites — 58 documents (8.3%), CC-BY-4.0:** generated by grounding an LLM in real, open-access CGIAR CGSpace publications (source_url retained for every grounded document, e.g. hdl.handle.net/10568/...) and rewriting them into the same factsheet format, so the corpus is not purely synthetic and includes traceable links back to genuine agricultural research.

Queries and relevance judgments were generated alongside the corpus (see Labelling Strategy) rather than collected from real farmer interactions, chat logs, or extension call-centre records. No real farmer questions, names, locations, or any other personal data were collected or used.

6. Web Collection Ethics

- **Consent / copyright:** the only externally sourced content is the 58 LLM-grounded documents drawn from CGIAR's CGSpace repository, an institutional open-access archive explicitly licensed CC-BY-4.0 (a small number of individual records carry CC-BY-SA-4.0, CC0-1.0, or CC-BY-3.0). Attribution is preserved for every grounded document via the *source*, *source_url*, and *license* fields, satisfying the attribution requirement of CC-BY licensing.
- **Privacy:** no personal, farmer-identifying, or web-scraped social data is present anywhere in the dataset. All content is institutional agronomic advisory material (crop, pest, disease, and climate guidance) at the fact-sheet level, not individual records about people.
- **No scraping of live/unlicensed web sources:** the dataset avoids ad-hoc web scraping; the only non-synthetic content comes from one clearly licensed, citable institutional source.

7. Bias and Representation

Where representation is strong

The corpus spans 21 African countries and 13 staple/commercial crops (maize 128 docs, tomato 102, rice 78, cassava 71, common bean 49, cowpea 42, groundnut 41, sorghum 36, plantain 31, yam 23, cocoa 21, pearl millet 10, plus 63 general climate/soil documents), which gives reasonable breadth across West, East, and Central African farming systems.

Where representation is uneven

- **Country skew:** Nigeria (133 docs), Ghana (103), Tanzania (92), and Kenya (75) dominate the corpus, while countries such as Togo (3), Benin (4), and Rwanda (6) have very thin coverage — a retriever trained or evaluated on this data will be systematically weaker for farmers in the low-coverage countries.
- **Crop skew:** four staple crops (maize, tomato, rice, cassava) account for over half the corpus; livestock, aquaculture, and many minor/orphan crops important to smallholder livelihoods are absent entirely.
- **Language:** every query and document is in English. Farmers who primarily communicate in local languages, pidgin, or via voice rather than text are not represented in the query style, even though they are a core part of the population the project aims to serve.
- **Source skew toward synthetic content:** because 91.7% of documents are synthetically authored rather than sourced from real extension publications, the dataset is a strong benchmark for retrieval mechanics but is not itself a validated source of agronomic ground truth.

How the team plans to improve this

- Prioritize additional documents and queries for the thin-coverage countries (Togo, Benin, Rwanda, Niger, Malawi) before treating retrieval performance there as reliable.
- Extend crop coverage to include livestock and additional minor/orphan crops in later iterations.
- Add paraphrase templates that reflect informal, pidgin, or voice-transcribed phrasing rather than only well-formed written English questions.

- Route any synthetic factsheet content through agronomic expert review before it is used to generate advice shown to a real farmer, treating this dataset as a retrieval-training resource rather than a validated knowledge source on its own.

8. Values and Community Alignment

The project's Challenge 1 problem statement centers on the gap between how few trained extension officers are available and how many smallholder farmers need timely, trustworthy agronomic guidance. This dataset operationalizes that value directly: it is built entirely around the plain-language questions a farmer would actually ask (not keyword search queries), and it rewards retrieval systems for surfacing the single most useful piece of institutional advice first (nDCG@5) rather than a generic list of loosely related documents. Grounding a meaningful share of the corpus in real CGIAR research (with full attribution) keeps the retrieval system tethered to credible agricultural science rather than purely synthetic content.

Accountability to the intended community depends on closing the coverage gaps identified above — particularly the country and crop skew — before the retrieval system's outputs are treated as reliable advice for farmers outside the best-covered regions and crops. The team considers this dataset a strong foundation for the retrieval component of the RAG pipeline, but not, by itself, a substitute for local agronomic expert validation.

Data Card prepared for TRI AI Saturdays Cohort 10 — Challenge 2 (Create a Data Card for the Dataset). Source: agricultural-extension-rag-smart-retrieval-for-farmers dataset, Kaggle.